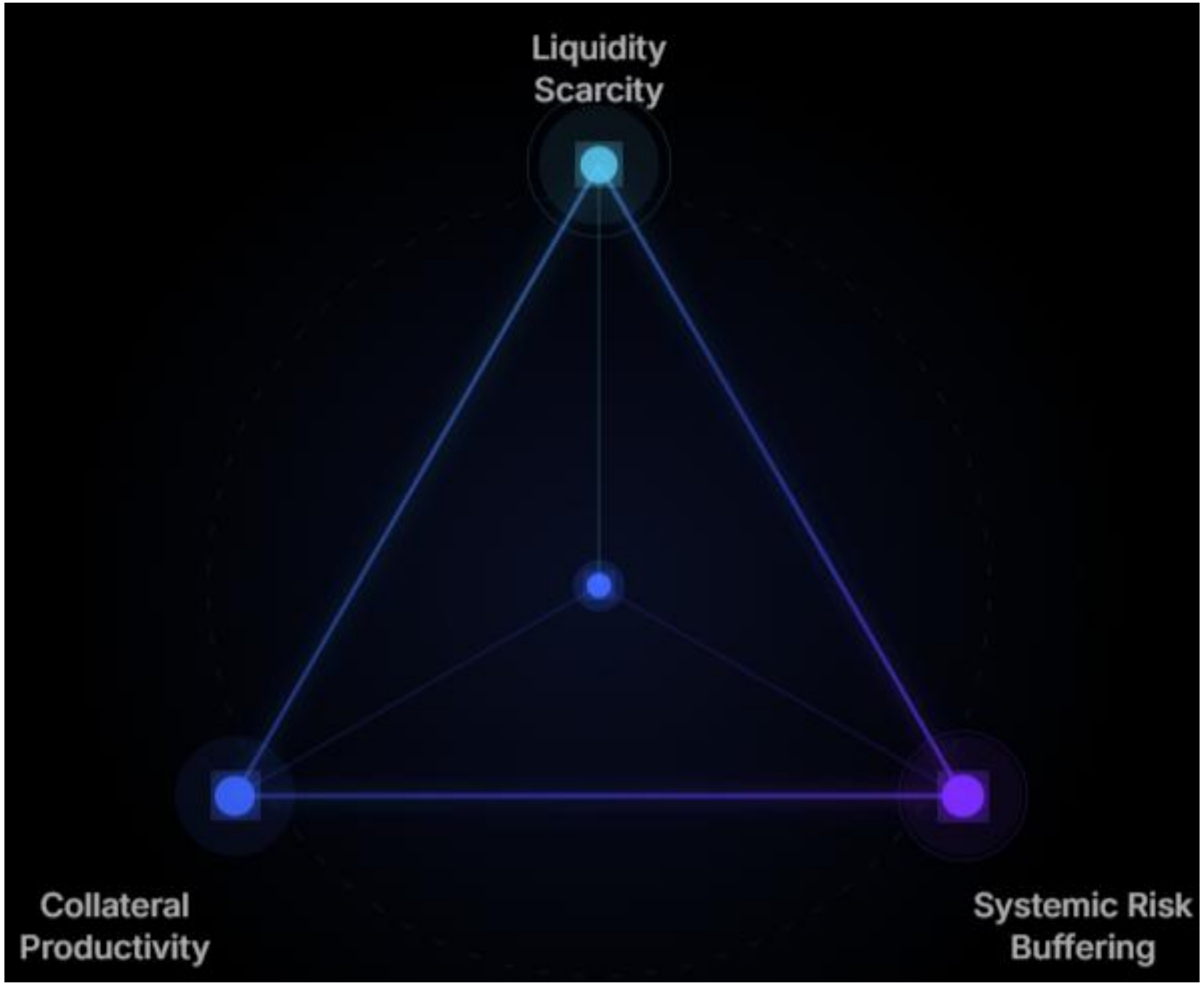
Martingale Protocol

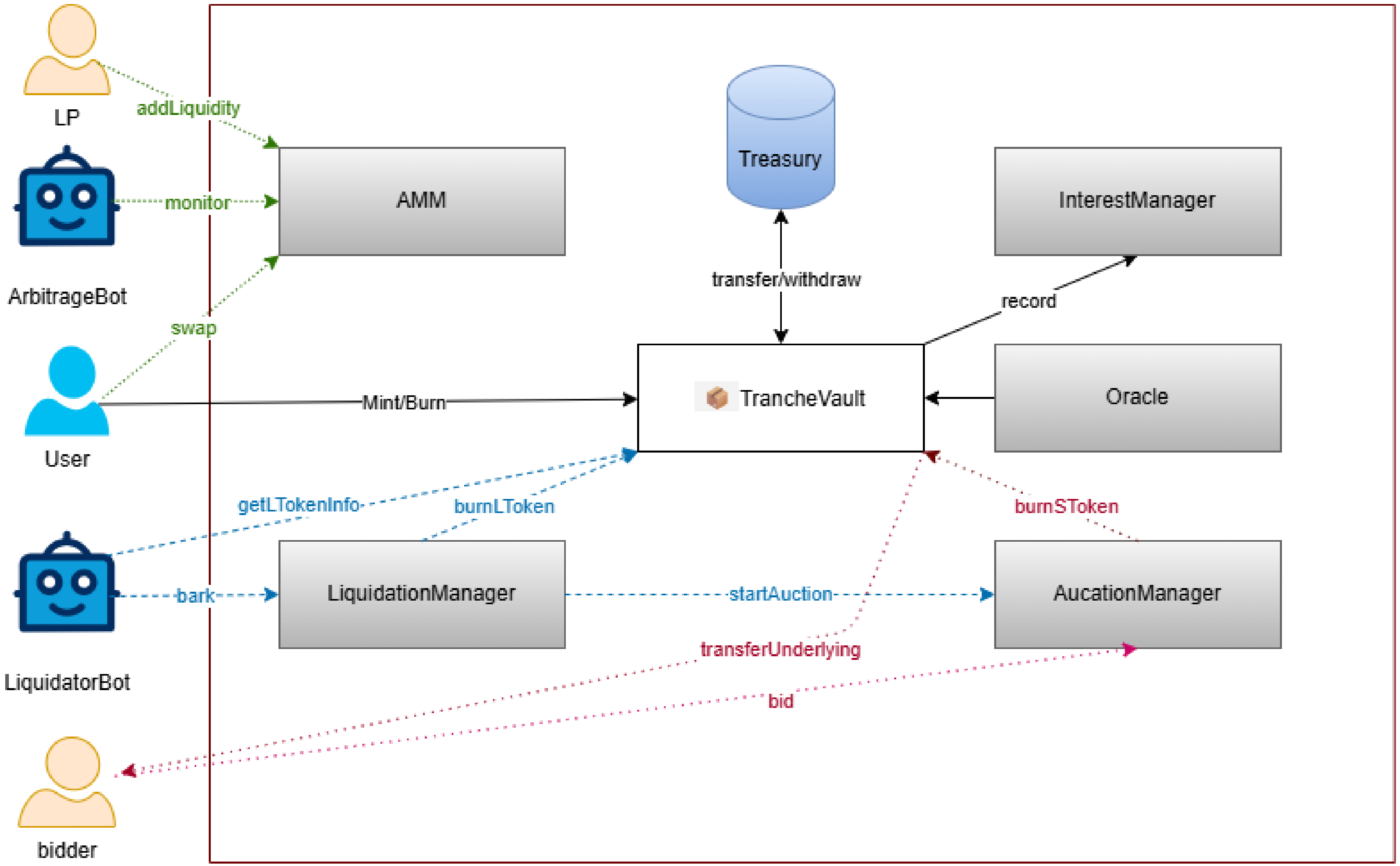
Martingale

Overview

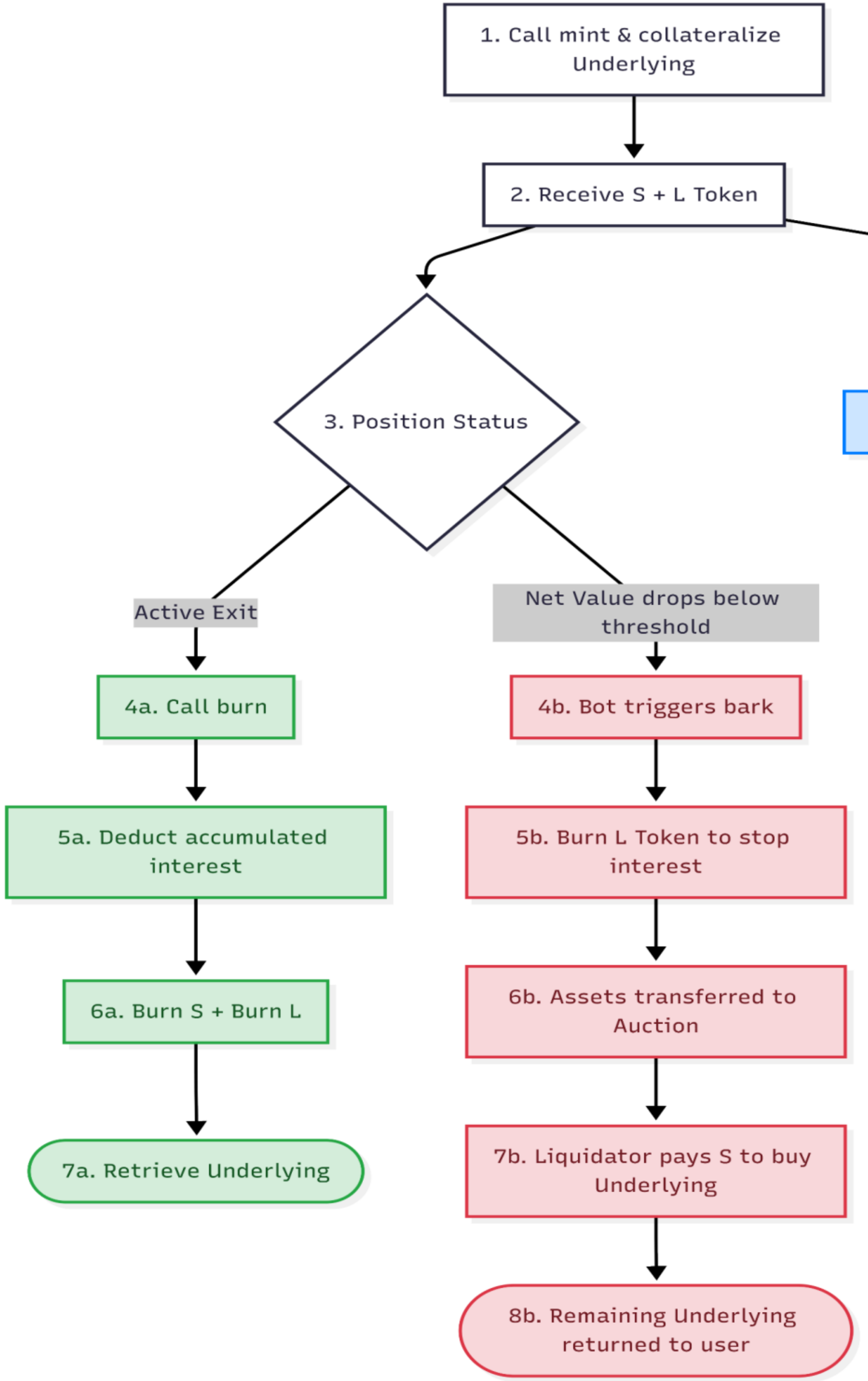
- Martingale protocol is decentralized credit, which combines stablecoin and lending Markets.
    - ✓ *Pledge underlying to borrow stable coins*
    - ✓ *Customized DeFi investment tools for users with different risk preferences*



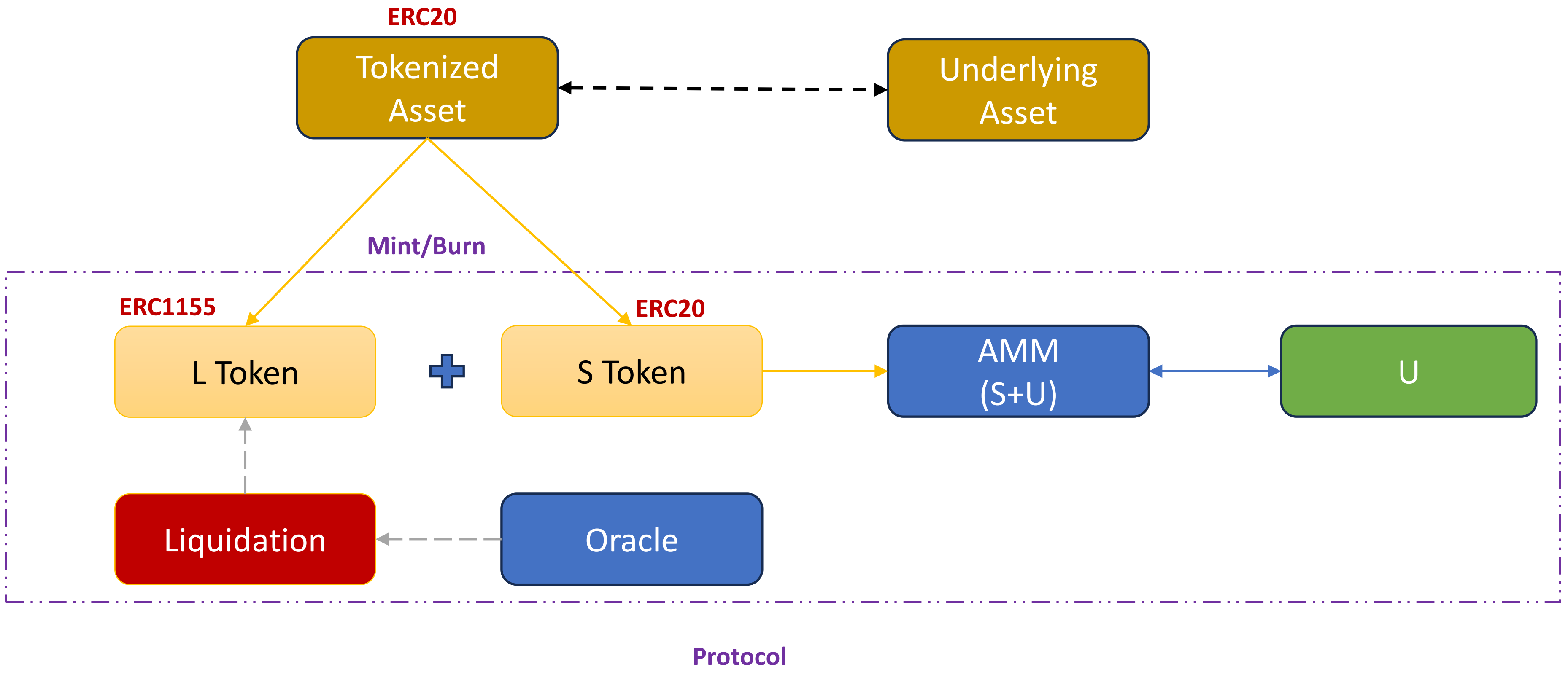
System Architecture



Life Cycle


1. Call mint & collateralize Underlying
2. Receive S + L Token
3. Position Status
Active Exit
Net Value drops below threshold
4a. Call burn
5a. Deduct accumulated interest
6a. Burn S + Burn L
7a. Retrieve Underlying
4b. Bot triggers bark
5b. Burn L Token to stop interest
6b. Assets transferred to Auction
7b. Liquidator pays S to buy Underlying
8b. Remaining Underlying returned to user
4c. Swap S to U in AMM Pool
5c. Obtain U

Model Ⅰ——Mint/Burn



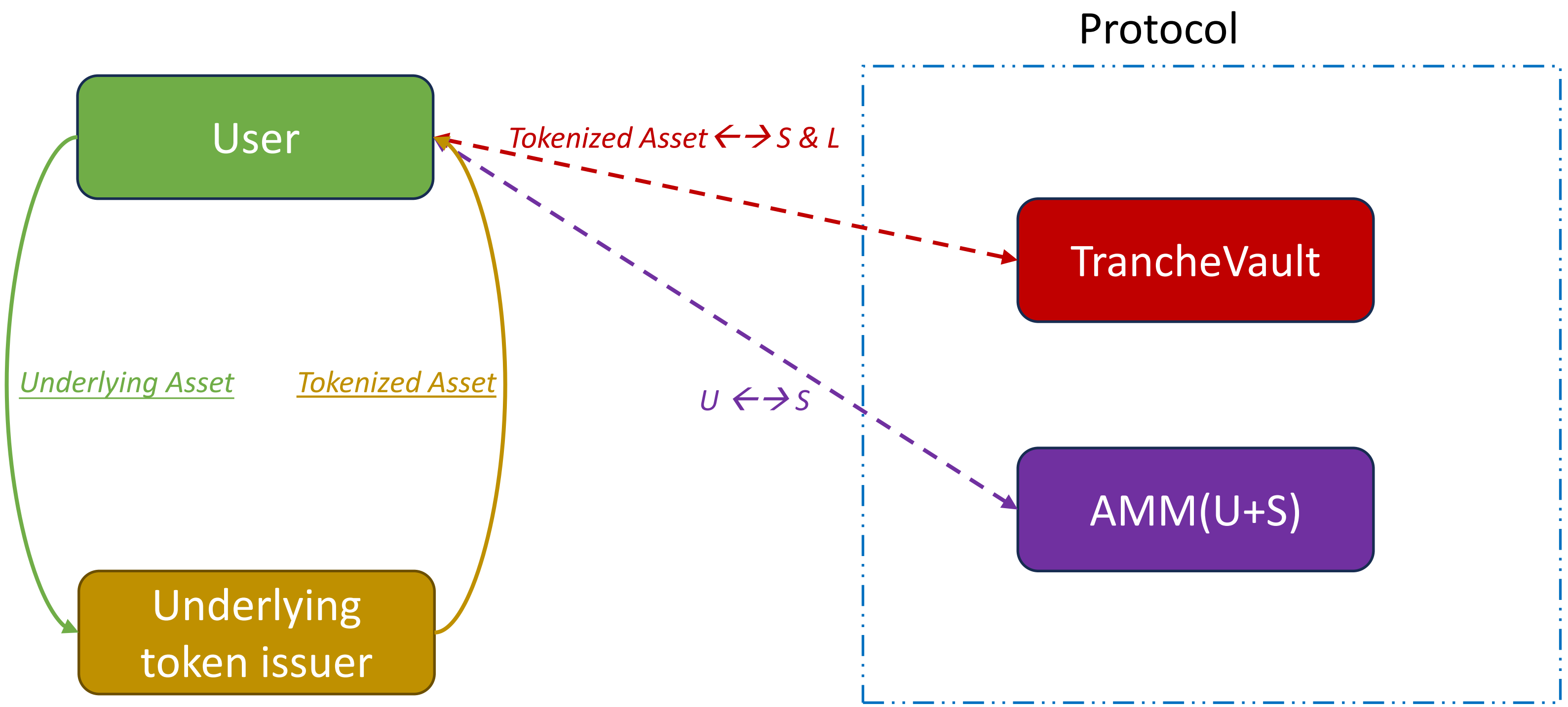
Model Ⅰ——Mint/Burn

- **Parameters**： $LTV_{\max} = 75\%$, $Liquidation = LTV_{\max} + x\%$
- **Mint**： Stake N_G Underlying ➔ Mint N_S stable token + N_L leverage token

t = 0	t > 0
$N_S * 1 + N_L * 1 = N_G * P_0$ $LTV_0 = \frac{N_S}{N_G * P_0}$ $NAV_0 = 1$	$N_S * 1 + N_L * NAV_t = N_G * P_t$ $LTV_t = \frac{N_S}{N_G * P_t}$ $NAV_t = \frac{P_t - P_0 * LTV_0}{P_0 - P_0 * LTV_0}$

- **Burn**： Users can burn minted assets in any proportion to redeem the underlying assets.

Model Ⅱ——Interest Rate Model



First lending: Lend out physical gold to receive tokens and earn interest.

Second lending: Pledge tokens to borrow USDT and pay interest.

Model Ⅱ——Interest Rate Model

Collateralize Tokenized Asset→Borrow U

Users pay interest to protocol

Model:

Gross Borrow Rate = Base Rate + Utilization Factor + Collateral Risk Premium

Base Rate: Lending interest rate for USD (U)

Utilization Factor: The fewer U in the AMM, the higher the factor

Collateral Risk Premium: Risk premium of the underlying asset

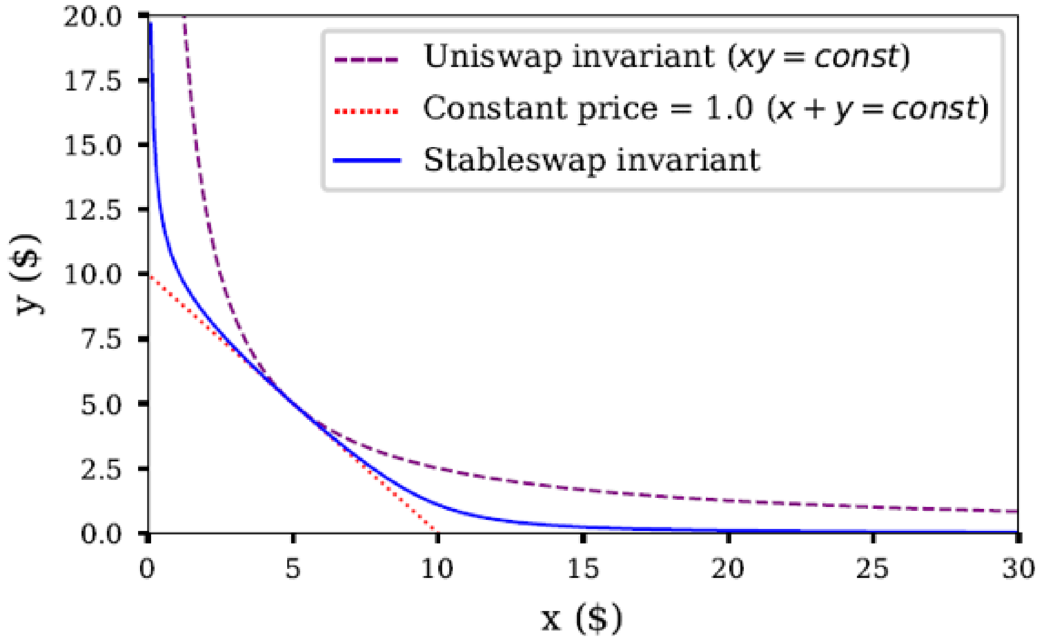
Practical:

Interest is accrued on a per-second basis and reflected in the net value of L tokens.

Upon burning, the deduction is made once in the form of the underlying asset

Model Ⅲ——AMM

- **AMM:**
  - S token + U
  - Stableswap invariant
- **Arbitrage:**
  - Use an arbitrage program to monitor AMM deviations.



Model Ⅳ——Oracle

Third-party oracles

Chainlink

Pyth Network

RedStone

Core features:

Delayed price updates

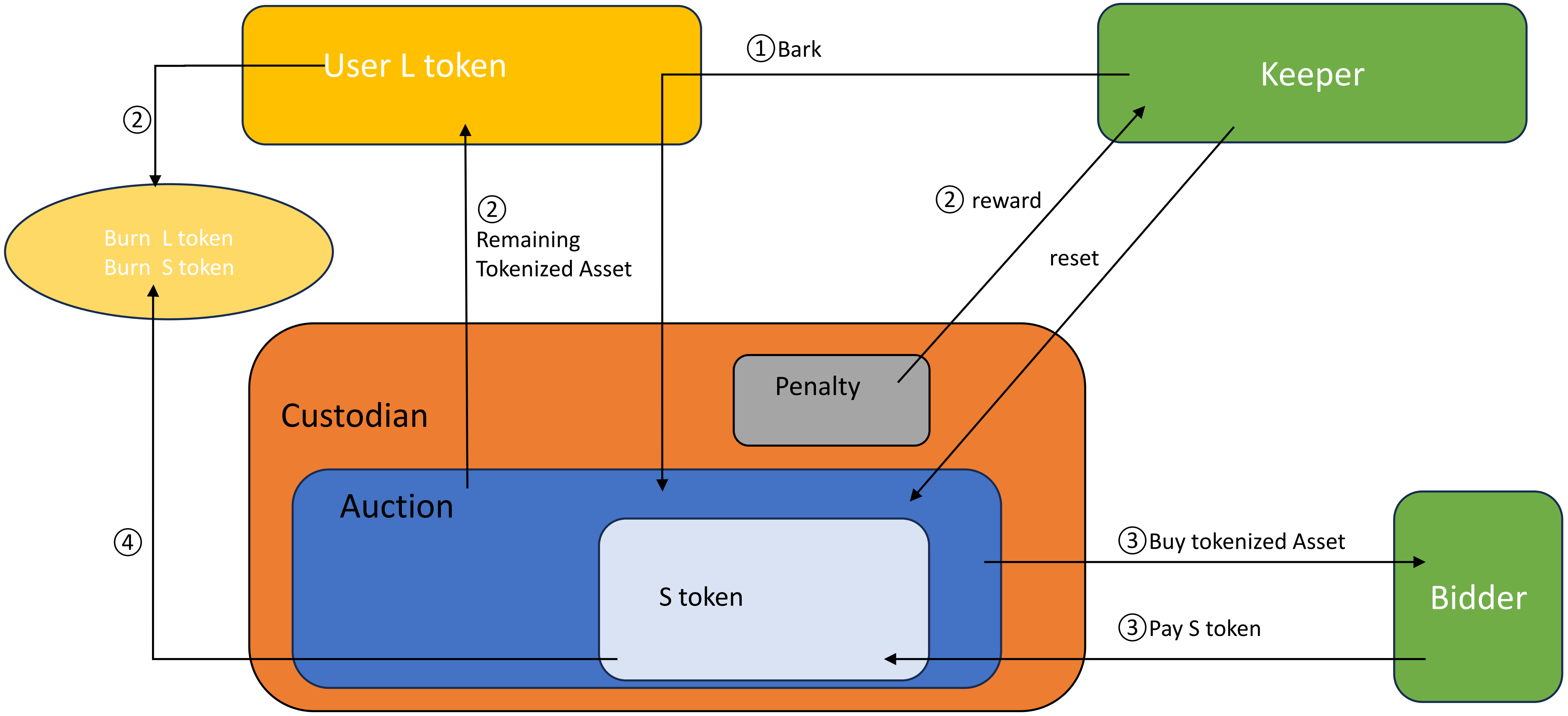
Multi-source aggregation

Multi-signature voting mechanism

Emergency halt

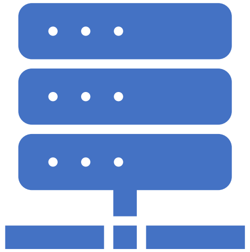
Price change threshold

Model Ⅴ——Liquidation



NOTE: The liquidation model determines the optimal liquidation boundary based on system parameters for users.

Mechanism



On-chain

Access Control / Multi-sig

Whitelist

Proxy Pattern

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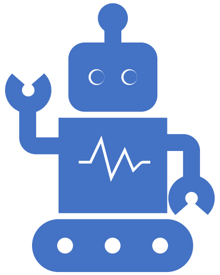

Off-chain

Liquidation bots

Arbitrage bots

Maintenances bots

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Website

Martingale: https://martingaleco.com

App Demo: https://test-al2.pages.dev/app

Documentation: https://test2026.gitbook.io/test-tranche_protocol/fMFG0viXW9hbW0MeCbeh/